

CAREERPATH AI

AI-Powered Career Guidance & Skill Development Tracker

PHASE 02 INTERNSHIP PROJECT REPORT

Submitted By
Aishwarya N R

Internship Project Submission

ABSTRACT

CareerPath AI is a MERN-based career guidance and skill-development platform designed to convert a student's career interests, skills, academic context and learning activity into an actionable development roadmap. Phase 02 extends the earlier career-assessment concept with explainable career recommendations, current-versus-target skill analysis, curated learning resources, goal tracking, progress analytics, an AI career assistant and a printable mentor-ready report.

The system follows a modular MERN architecture. The React frontend presents the student workspace, while an Express backend exposes REST APIs for user access, recommendations, skill-gap analysis, learning, goals, analytics, chat and reporting. MongoDB is supported through Mongoose, while a demo fallback mode allows the application to be demonstrated without a database. The recommendation layer is intentionally explainable: career profiles are compared with the student's skills and interests, and results are ranked using weighted overlap. An optional Python TF-IDF prototype and optional Gemini provider demonstrate paths toward more advanced AI capabilities.

TABLE OF CONTENTS

1. Introduction
2. Problem Statement
3. Objectives
4. Existing System
5. Proposed System
6. System Requirements
7. Technology Stack
8. System Architecture
9. Project Modules
10. Recommendation & Skill-Gap Method
11. Database Design
12. REST API Design
13. Authentication & Security
14. User Interface & Screens
15. Testing & Validation
16. Results
17. Advantages
18. Limitations
19. Future Enhancements
20. Conclusion
21. References

1. INTRODUCTION

CareerPath AI is a MERN-based career guidance and skill-development platform designed to convert a student's career interests, skills, academic context and learning activity into an actionable development roadmap. Phase 02 extends the earlier career-assessment concept with explainable career recommendations, current-versus-target skill analysis, curated learning resources, goal tracking, progress analytics, an AI career assistant and a printable mentor-ready report.

Phase 02 focuses on transforming career assessment results into a measurable skill-development experience. Instead of presenting only a career label, the platform provides a connected workflow: understand the target role, inspect skill gaps, follow learning resources, set goals, monitor analytics and export a development report.

2. PROBLEM STATEMENT

Students often receive career suggestions without a clear explanation of why a role is suitable, which skills are missing, what to learn next, or how progress should be measured. CareerPath AI addresses this gap by connecting career matching with skill-gap analysis, learning resources, goals and progress reporting in one interface.

3. OBJECTIVES

- Build a centralized student career-development workspace.
- Generate transparent and explainable career match scores from skills, interests and target-career context.
- Compare current proficiency against target proficiency for a selected career.
- Prioritize the most important skill gaps and recommend the next learning action.
- Provide curated learning resources with completion and progress tracking.
- Allow students to create and monitor career-development goals.
- Visualize readiness, skill coverage, learning velocity and growth trends.
- Provide an AI career assistant for practical questions on careers, skills, projects and interviews.
- Generate a professional progress report that can be shared with an internship mentor.

4. EXISTING SYSTEM

Traditional career guidance tools commonly provide static career descriptions, generic course lists or isolated assessment results. Such approaches can make it difficult for students to understand the relationship between their current skills and the requirements of a target role. Progress is also often tracked separately from recommendation results.

5. PROPOSED SYSTEM

CareerPath AI integrates career recommendation, skill-gap analysis, learning resources, goals, analytics and AI-assisted guidance into a single workspace. The proposed flow is:

- Student context → Career matching → Skill-gap identification → Priority learning → Goal tracking → Analytics → Mentor-ready report

Core implemented features:

Module	Purpose
Secure access	Login, protected routes, local session persistence and JWT-ready authentication.
Personalized dashboard	Readiness score, learning streak, completed courses, learning hours and current roadmap.
AI career recommendations	Ranked career paths with match percentages, skill tags and explanation/reason fields.
Skill gap analysis	Current vs target proficiency for Python, SQL, Statistics, Machine Learning, Visualization and Deployment.
Learning Hub	Curated courses/videos/certification resources, progress bars, completion actions and external links.
Goals & Progress	Career goals with category, due date, progress percentage and new-goal creation.
Career AI	Chat interface with rule-based guidance and optional Gemini integration.
Analytics	Skill proficiency chart, readiness trend, learning metrics and export/report navigation.
Performance report	Print/save-to-PDF report with executive summary, skills, recommendation and next steps.
Profile	Student identity, education, target career, focus areas and current toolkit.

6. SYSTEM REQUIREMENTS

Category	Requirement
Operating system	Windows / Linux / macOS
Frontend runtime	Node.js with Vite-based React development
Backend runtime	Node.js + Express
Database	MongoDB / MongoDB Atlas for persistent mode; optional for demo
Browser	Modern Chromium/Firefox/Safari browser
Development tools	VS Code or equivalent IDE, npm
Optional AI	Gemini API key when AI_PROVIDER=gemini
Frontend URL	http://localhost:5173
Backend URL	http://localhost:5000

7. TECHNOLOGY STACK

Layer	Technology / Implementation
Frontend	React 18, Vite, React Router, Axios, Recharts, Lucide React
Backend	Node.js, Express 5, REST APIs, CORS, dotenv
Database	MongoDB with Mongoose; demo fallback mode works without MongoDB
Authentication	JWT-ready authentication, bcrypt password hashing, protected routes
AI / Recommendation	Explainable weighted-overlap recommendation engine; optional TF-IDF/scikit-learn prototype and optional Gemini provider
Visualization	Recharts for bar, line and area charts
Development	VS Code/local development, frontend on localhost:5173, API on localhost:5000

8. SYSTEM ARCHITECTURE

The application is organized into presentation, API, logic, persistence and optional AI layers.

Layer	Implementation
Presentation Layer	React UI, reusable components, routing, responsive dashboard screens
Service/API Layer	Express REST endpoints for authentication, recommendations, skills, learning, goals, analytics, chat and reports
Logic Layer	Recommendation engine, skill-gap calculation and career-chat service
Persistence Layer	MongoDB/Mongoose models for users and goals; demo fallback data for presentation mode
Optional AI Layer	Gemini integration when AI_PROVIDER=gemini and an API key are configured

High-level request flow:

- React page
- Axios API request
- Express route
- Authentication middleware
- Business logic / recommendation service
- MongoDB or demo data
- JSON response
- React dashboard update

9. PROJECT MODULES

1. Authentication & Access Control

The login screen supports credential-based access and an instant demo workspace. ProtectedRoute prevents access to application pages when no authenticated user is present. Auth state is persisted in local storage for the demo experience.

2. Dashboard

The dashboard provides a personalized snapshot: career readiness, learning streak, completed courses, learning hours, weekly learning momentum, current Data Science roadmap and top career matches.

3. AI Recommendations

The recommendation engine evaluates overlap between a user's skills/interests and predefined career profiles. A target-career boost is applied when the user's target matches a career. Results are ranked and displayed with reasons, skill tags and match percentages.

4. Skill Gap Analysis

The skill matrix compares current and target levels. The interface labels skills as Strength, Almost there or Priority gap and highlights the next best action. The implemented demo uses six tracked skills.

5. Learning Hub

Learning resources are linked to high-priority skill gaps. Each item shows type, provider, skill, duration, progress and an action to mark completion or open the resource.

6. Goals & Progress

Students can view active goals, progress and due dates and can create a new personal goal from the interface.

7. Career AI

The chat interface provides contextual career guidance. The backend can use Gemini when configured; otherwise a practical rule-based response service provides fallback guidance.

8. Analytics

Analytics summarizes readiness, learning velocity, skill coverage and consistency. Recharts visualizations show current proficiency and readiness growth over time.

9. Performance Report

The report module turns platform metrics into a structured development summary and provides Print / Save PDF functionality for mentor sharing.

10. Profile

The profile module displays student details, target career, education, focus areas and the current technical toolkit.

10. RECOMMENDATION & SKILL-GAP METHOD

10.1 Explainable Career Recommendation

The live JavaScript recommendation engine stores six career profiles. For each career, the system calculates skill overlap and interest overlap between the student profile and the career profile. The implemented scoring formula combines 58% skill overlap, 30% interest overlap, an optional 12-point target-career boost, and a base offset of 8, with the final value capped at 98.

Conceptual formula:

$$\text{Match Score} = 0.58 \times \text{SkillOverlap} + 0.30 \times \text{InterestOverlap} + \text{TargetBoost} + 8$$

This approach is useful for an internship project because the recommendation can be explained to the student instead of behaving like a black box. The source code also includes an optional Python TF-IDF/cosine-similarity prototype for text-based career matching.

10.2 Skill-Gap Analysis

The skill-gap service evaluates six target skills: Python, SQL, Statistics, Machine Learning, Data Visualization and Model Deployment. Each skill contains a current score and a target score. The interface classifies a skill as a strength when current proficiency reaches the target, near when it is within 12 points of the target, or a gap otherwise.

Skill	Current	Target	Gap
Python	86%	85%	0
SQL	78%	80%	2
Statistics	54%	82%	28
Machine Learning	61%	84%	23
Data Visualization	68%	80%	12
Model Deployment	38%	70%	32

11. DATABASE DESIGN

MongoDB persistence is implemented through Mongoose models. The current project defines User and Goal models. Learning resources, analytics and report values are currently supplied from application services/demo data, making the platform easy to demonstrate even when MongoDB is unavailable.

Collection / Model	Important Fields	Purpose
User	name, email, password, targetCareer, skills, interests, timestamps	Stores student identity and career context.
Goal	userId, title, category, due, progress, status, timestamps	Stores student development goals and progress.

12. REST API DESIGN

Endpoint	Method	Purpose
/api/health	GET	System health and phase status
/api/auth/login	POST	Authenticate user / demo account
/api/auth/register	POST	Create user account
/api/auth/me	GET	Return authenticated user context
/api/recommendations	GET	Return ranked career recommendations
/api/recommendations/skill-gap	GET	Return target role, readiness and skill matrix
/api/learning	GET	Return curated learning resources
/api/learning/:id	PATCH	Update learning progress
/api/goals	GET	Return user goals
/api/goals	POST	Create a goal
/api/analytics/overview	GET	Return progress metrics and weekly activity
/api/chat	POST	Return Career AI guidance
/api/report	GET	Return mentor-ready report metrics

13. AUTHENTICATION & SECURITY

- Protected routes are implemented through a React ProtectedRoute component.
- The backend includes JWT verification middleware for authenticated API access.
- Passwords are hashed with bcryptjs when persistent registration is used.
- CORS is configured for the local React frontend.
- Environment variables are used for MongoDB URI, JWT secret and optional Gemini configuration.
- The demo mode intentionally uses a non-production demo token and should not be used as a production authentication strategy.

14. USER INTERFACE & SCREENS

The interface follows a consistent dashboard design language with a dark navigation rail, light content canvas, purple accent system, progress indicators, metric cards, charts and responsive panels. The following screenshots document the implemented Phase 02 user experience.

Figure 1. Login & Demo Access

Split-screen authentication experience with CareerPath AI branding, Phase 02 messaging and demo workspace access.

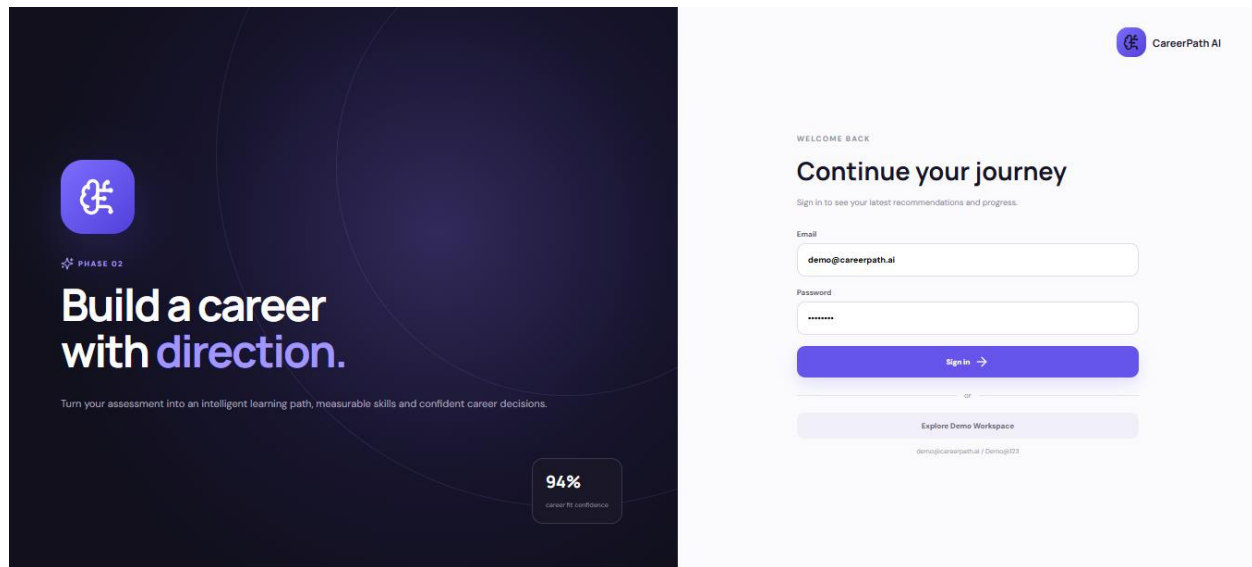


Figure 2. Personalized Dashboard

Overview of readiness, learning streak, completed courses, learning hours, momentum and current Data Science roadmap.

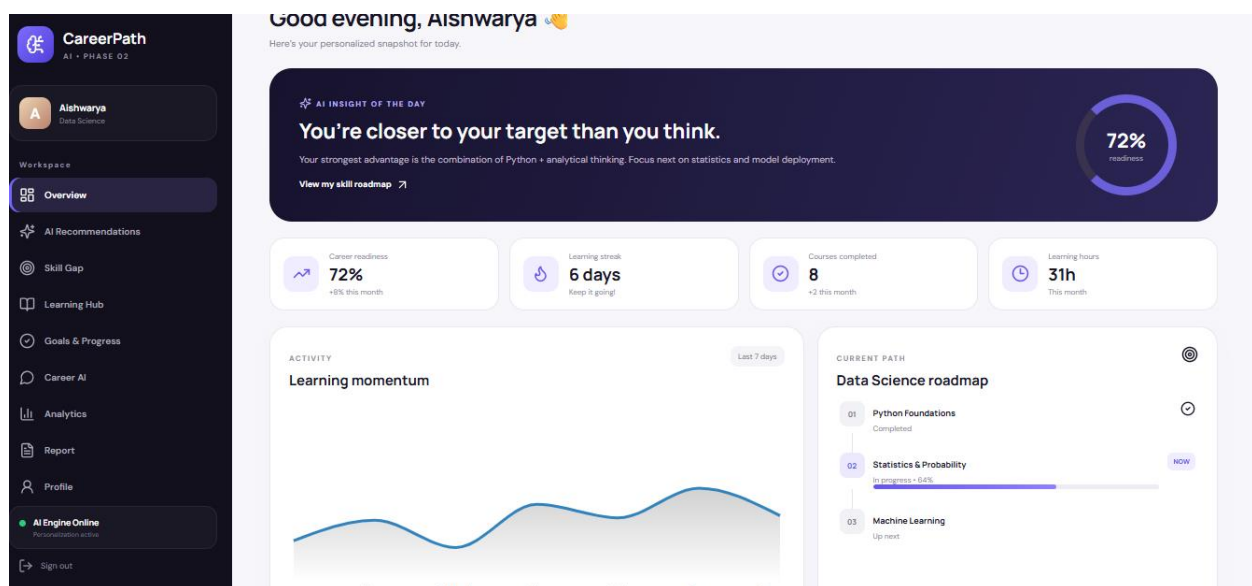


Figure 3. AI Career Recommendations

Explainable recommendation view with ranked career paths and match scores.

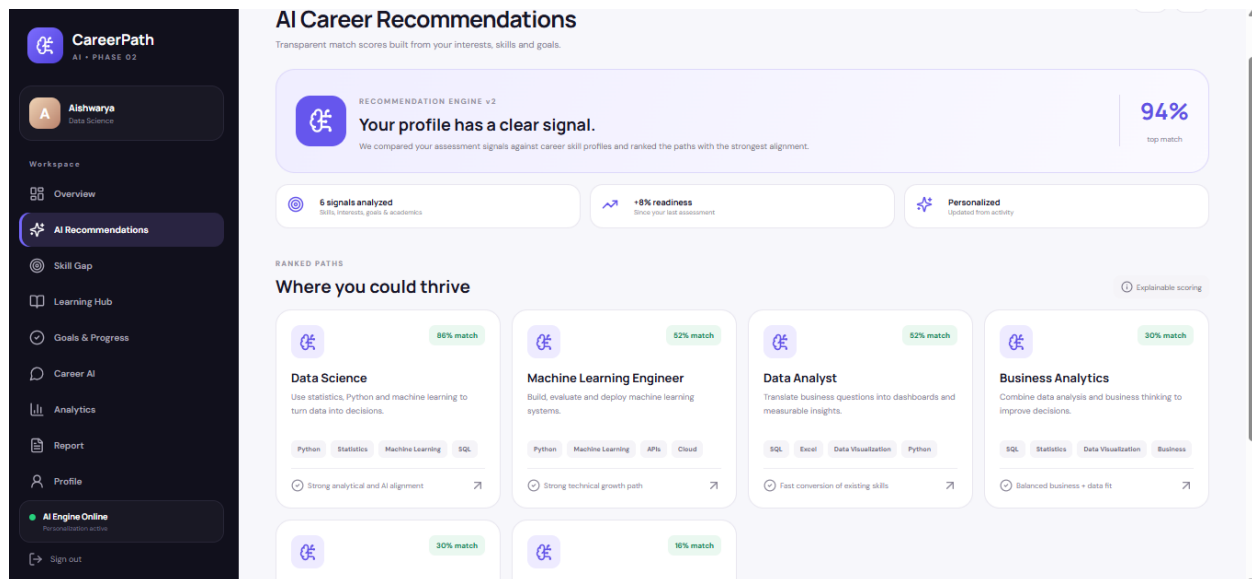


Figure 4. Skill Gap Analysis

Current-versus-target proficiency matrix with priority gaps and next-best action.

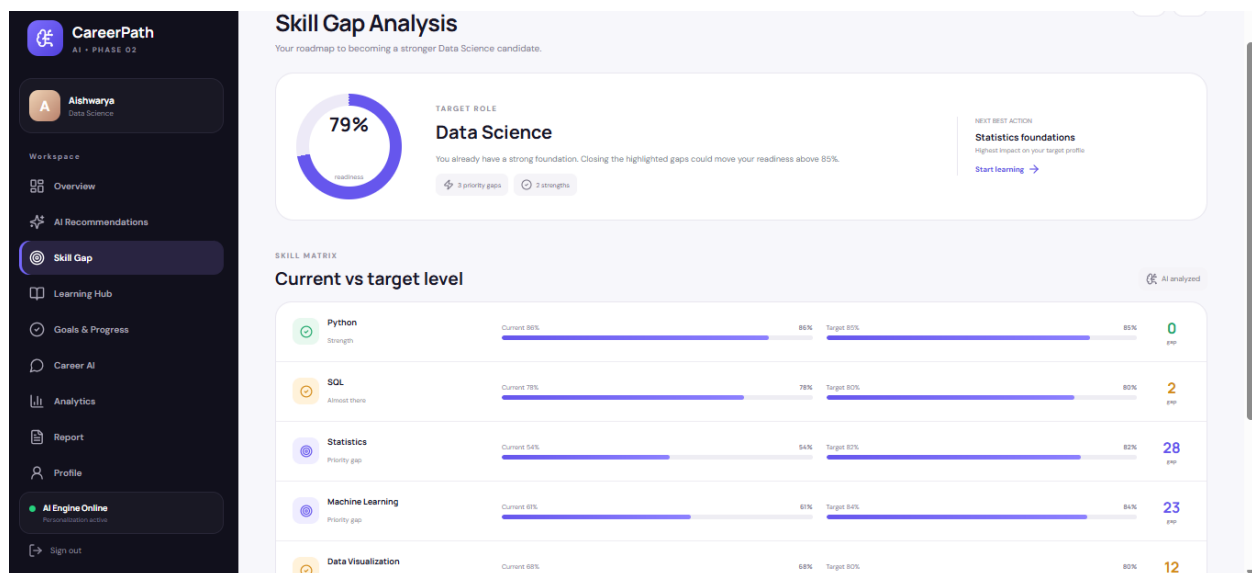


Figure 5. Learning Hub

AI-curated learning resources with providers, durations, progress and completion controls.

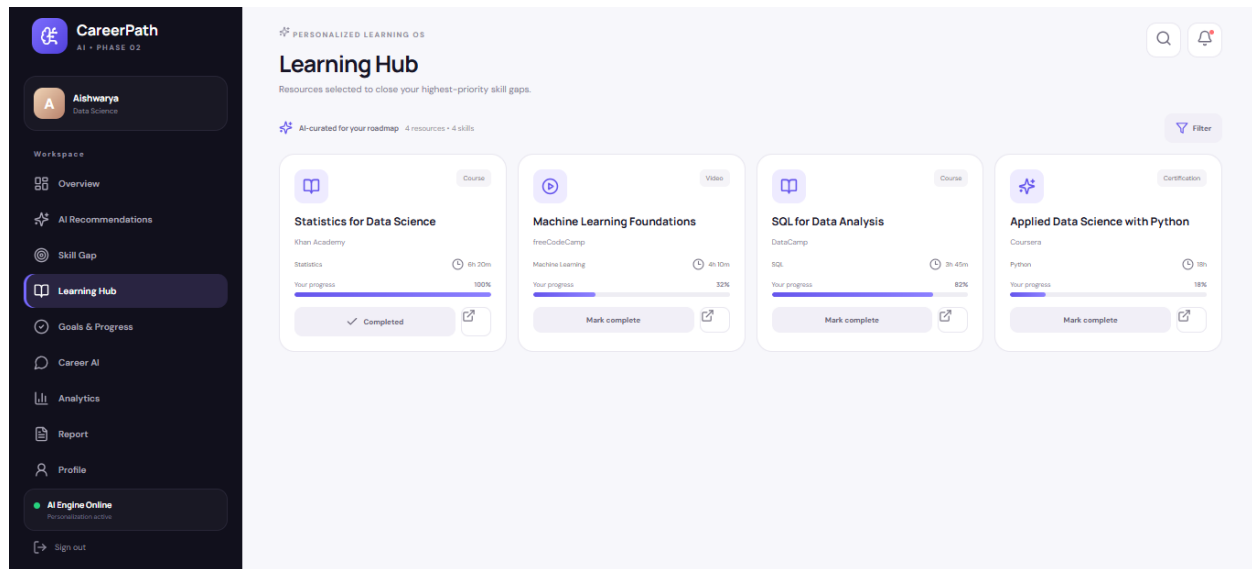


Figure 6. Goals & Progress

Goal summary, active goals, due dates and progress bars.

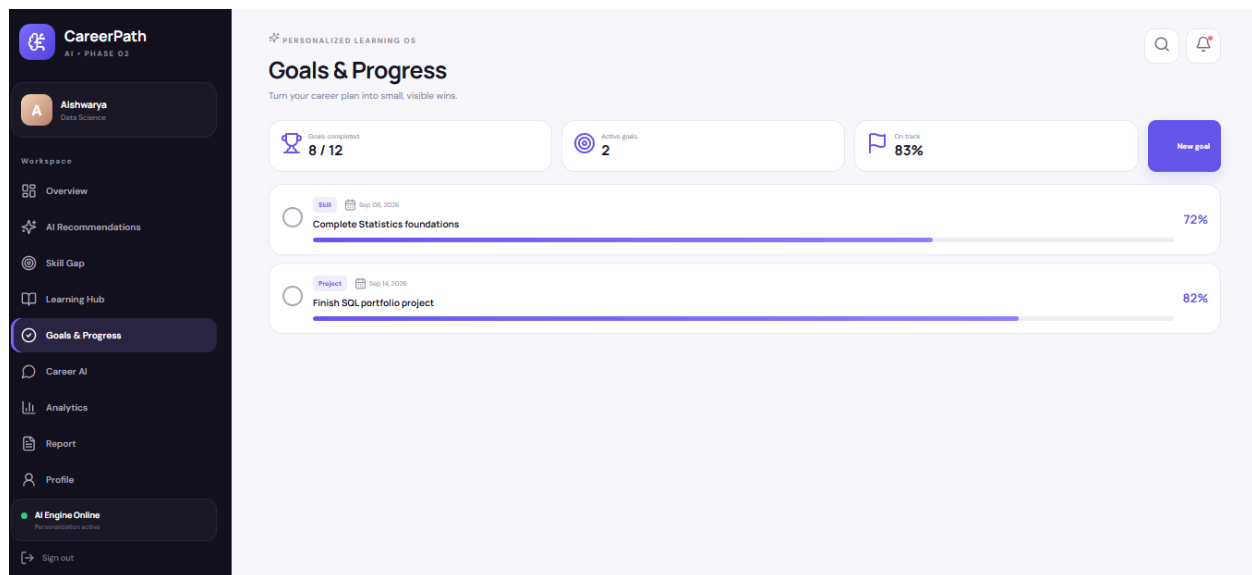


Figure 7. Career AI Assistant

Conversational career guidance interface with contextual AI information.

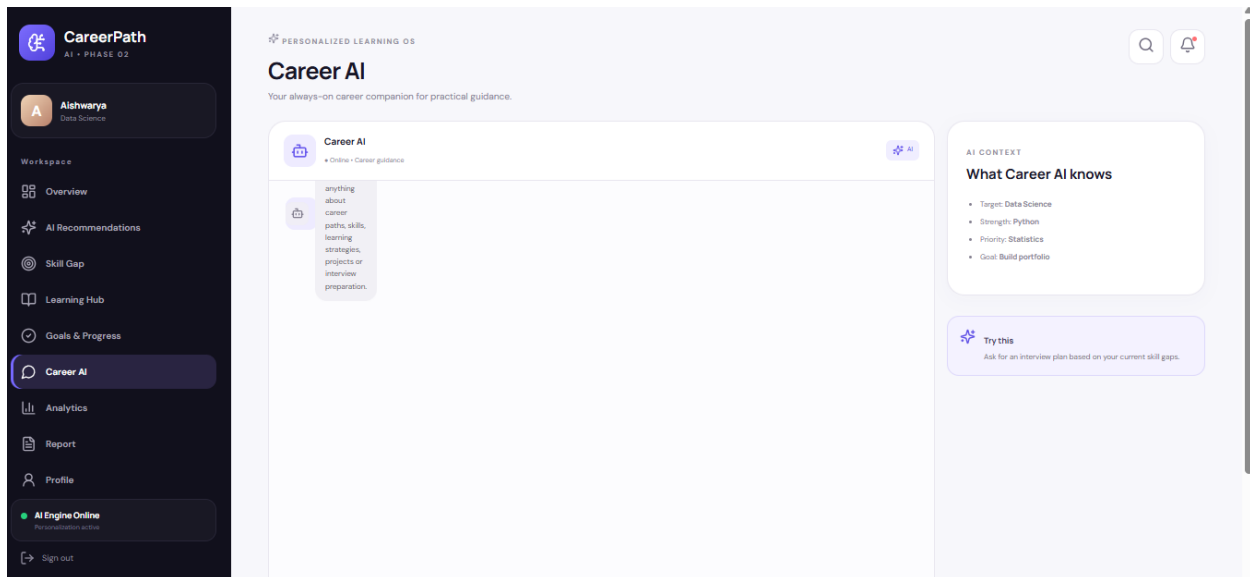


Figure 8. Progress Analytics

Skill proficiency chart, readiness trend and learning performance indicators.

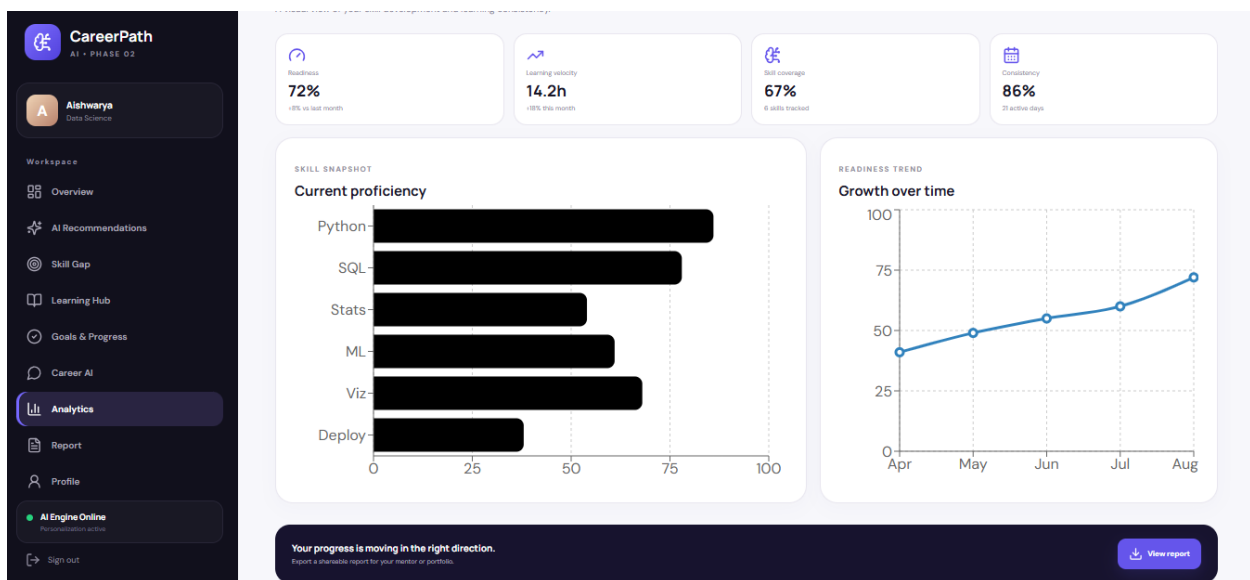


Figure 9. Mentor-Ready Development Report

Printable CareerPath Development Report with executive summary, skill profile and recommended next steps.

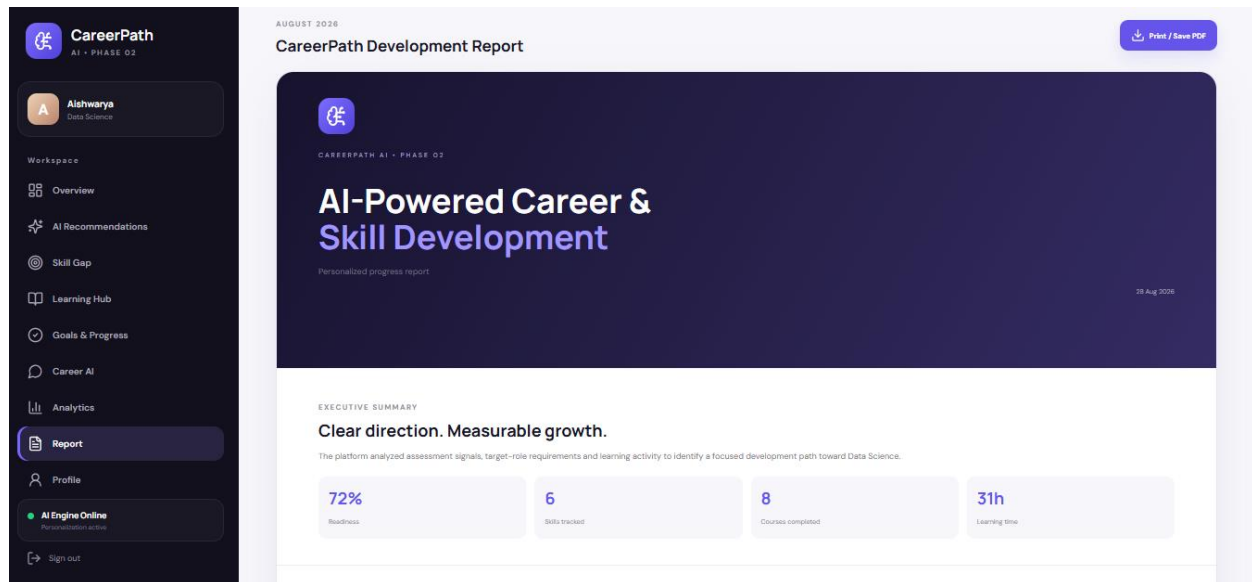
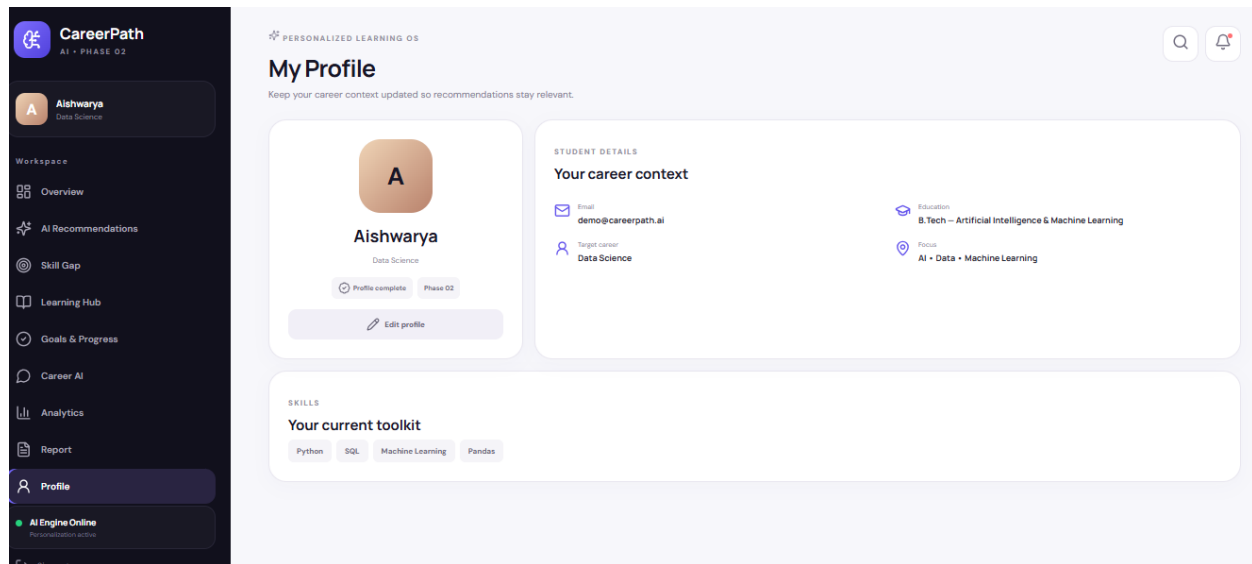


Figure 10. Student Profile
 Career context, education, target role and current technical toolkit.



15. TESTING & VALIDATION

Validation was performed against the implemented demo workspace and the supplied UI evidence. The table below summarizes the functional checks represented by the project and screenshots.

Test Case	Action	Expected Result	Status
Login / Demo access	Enter demo credentials or use demo workspace	Dashboard opens with personalized context	Pass
Protected routes	Open application route without authentication	Redirects to login	Pass
Recommendations	Open AI Recommendations	Ranked career cards with match scores and reasons	Pass

Skill gap	Open Skill Gap	Current vs target matrix and priority gaps shown	Pass
Learning	Open Learning Hub	Resources, progress and completion actions shown	Pass
Goals	Open Goals & Progress and create a goal	Goal form and progress cards available	Pass
Analytics	Open Analytics	Metrics plus charts rendered	Pass
Career AI	Submit a career question	Assistant response displayed in chat	Pass
Report	Open Report and use Print / Save PDF	Structured report view available	Pass
Profile	Open Profile	Student context and toolkit displayed	Pass

16. RESULTS

The completed Phase 02 prototype demonstrates an end-to-end career development workflow. In the supplied demo workspace, the interface shows a 72% readiness indicator, 6 tracked skills, 8 completed courses and 31 learning hours in the report view, with Data Science presented as the target direction and a 94% top career match. These values are application/demo metrics rather than independently measured real-world outcomes.

- Career recommendations are presented as ranked, explainable cards.
- Skill gaps are translated into a prioritized learning sequence.
- Learning progress, goals and analytics are presented in one dashboard.
- The Career AI interface provides contextual guidance with an optional Gemini path.
- A mentor-ready report can be printed or saved as PDF from the browser.

17. ADVANTAGES

- Personalized instead of purely generic career guidance.
- Explainable recommendation scores improve transparency.
- Connects recommendations directly to skill gaps and learning actions.
- Progress tracking encourages measurable, goal-oriented development.
- Analytics provide a visual summary of growth.
- Modular MERN architecture supports future expansion.
- Demo fallback mode simplifies internship presentation and evaluation.
- Professional report generation supports mentor review.

18. LIMITATIONS

- The default demo experience uses predefined profile, skill, learning and analytics values.
- The primary JavaScript recommendation engine is explainable rule-based scoring; the Python TF-IDF module is an optional prototype rather than the live scoring path.
- Gemini functionality is optional and requires configuration of AI_PROVIDER and GEMINI_API_KEY.
- Only User and Goal MongoDB models are currently defined; several dashboard datasets are intentionally demo-oriented.
- The learning filter currently demonstrates a lightweight filter interaction rather than a full multi-filter search system.

19. FUTURE ENHANCEMENTS

- Connect the recommendation engine to a larger career-skill ontology or labor-market dataset.
- Persist learning progress and goal updates per user in MongoDB rather than demo fallback state.
- Add richer assessment history and model-driven readiness prediction.
- Introduce role-specific learning-path generation using a production LLM with guardrails.
- Add resume parsing and project/portfolio evaluation.
- Add recruiter-oriented skill evidence and achievement badges.
- Deploy the application with HTTPS, production secrets, monitoring and CI/CD.
- Add automated unit, integration and end-to-end test suites.

20. CONCLUSION

CareerPath AI Phase 02 successfully extends a career-assessment concept into a practical AI-assisted skill development platform. The project combines a modern React interface with an Express REST backend, optional MongoDB persistence, explainable recommendation logic, skill-gap analysis, learning resources, goals, analytics, AI chat and report generation. The result is a mentor-ready prototype that demonstrates both software engineering implementation and product-oriented thinking. The architecture also provides a clear foundation for future improvements such as richer datasets, advanced machine-learning models, production LLM integration and full cloud deployment.